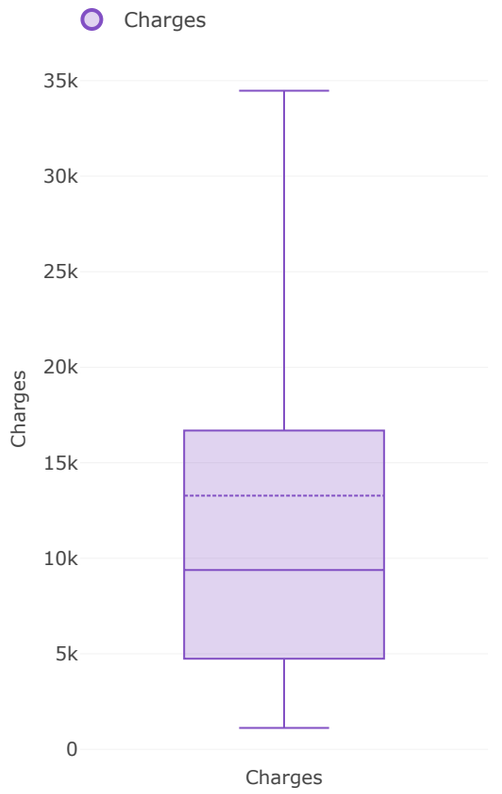
What factors have the strongest influence on medical insurance charges, and how do charges differ between smokers and non-smokers across age groups and regions?

1338
Number of Beneficiaries

$13,270.42
Average Charges

Region	Number
southeast	364
northwest	325
southwest	325
northeast	324
Total	**1338**

Descriptive Statistics of Charges



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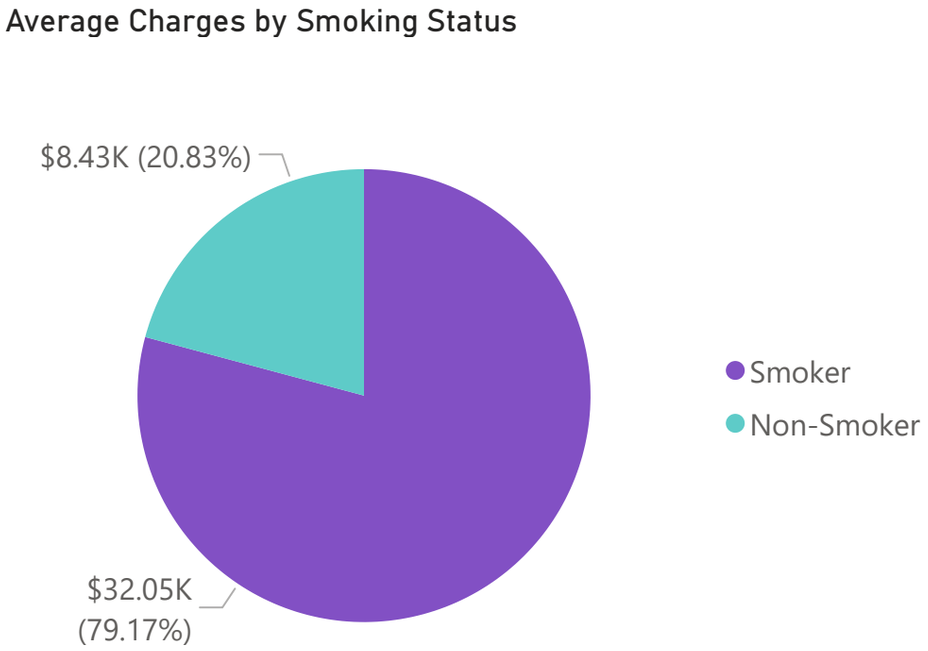
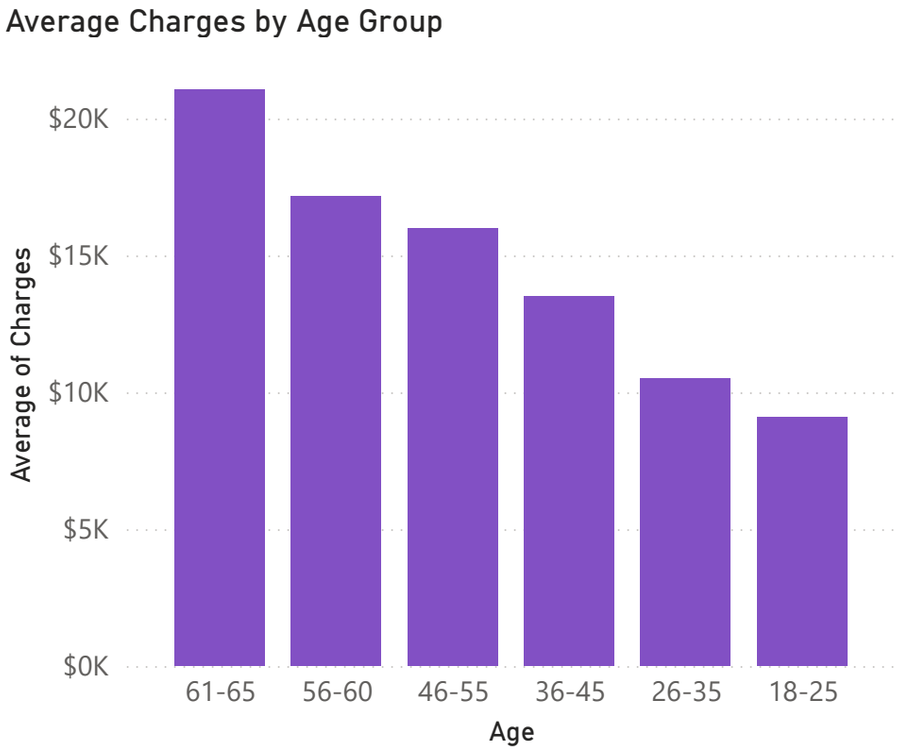
The Strongest Factors are Smoking and Age:





Smokers have **280%** higher medical insurance charges compared to non-smokers.

People aged 61-65 incur **65.83%** higher average insurance charges compared to the weighted average of charges of the other age groups.

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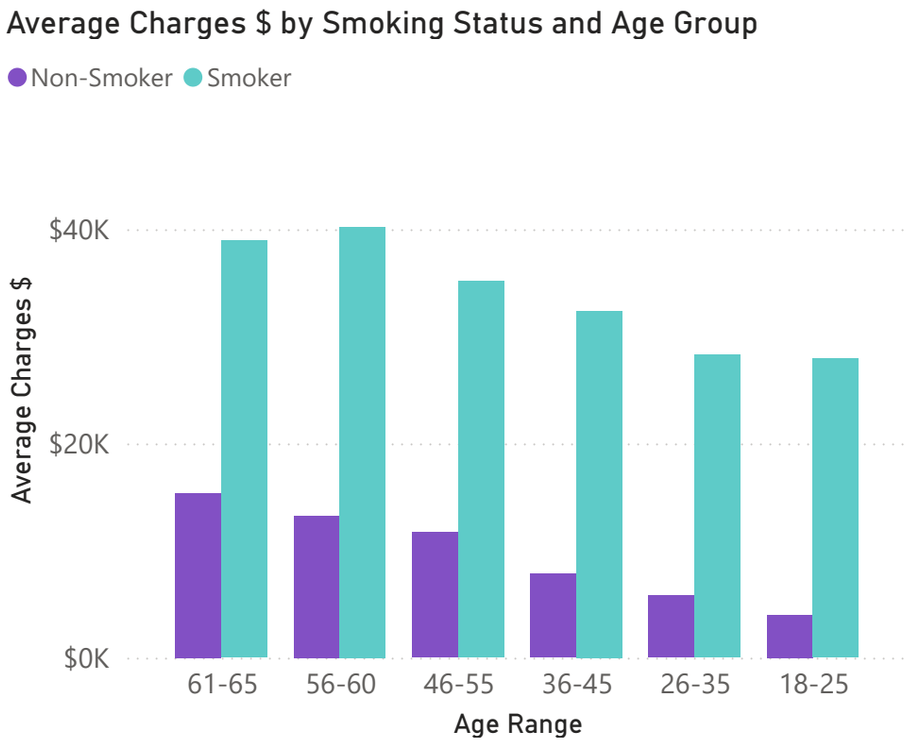
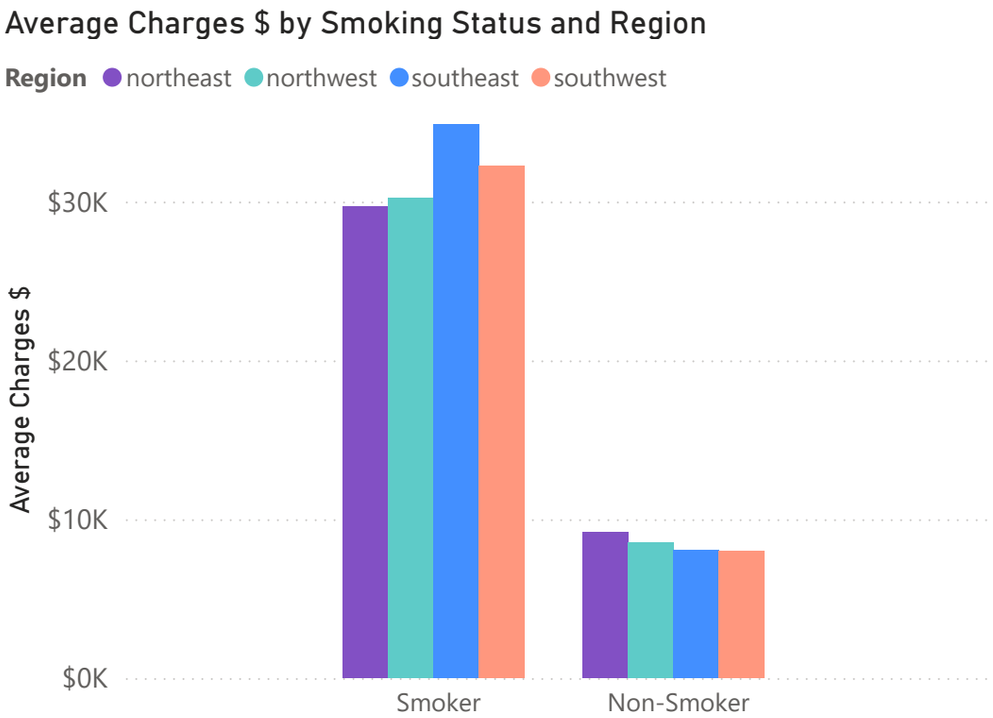
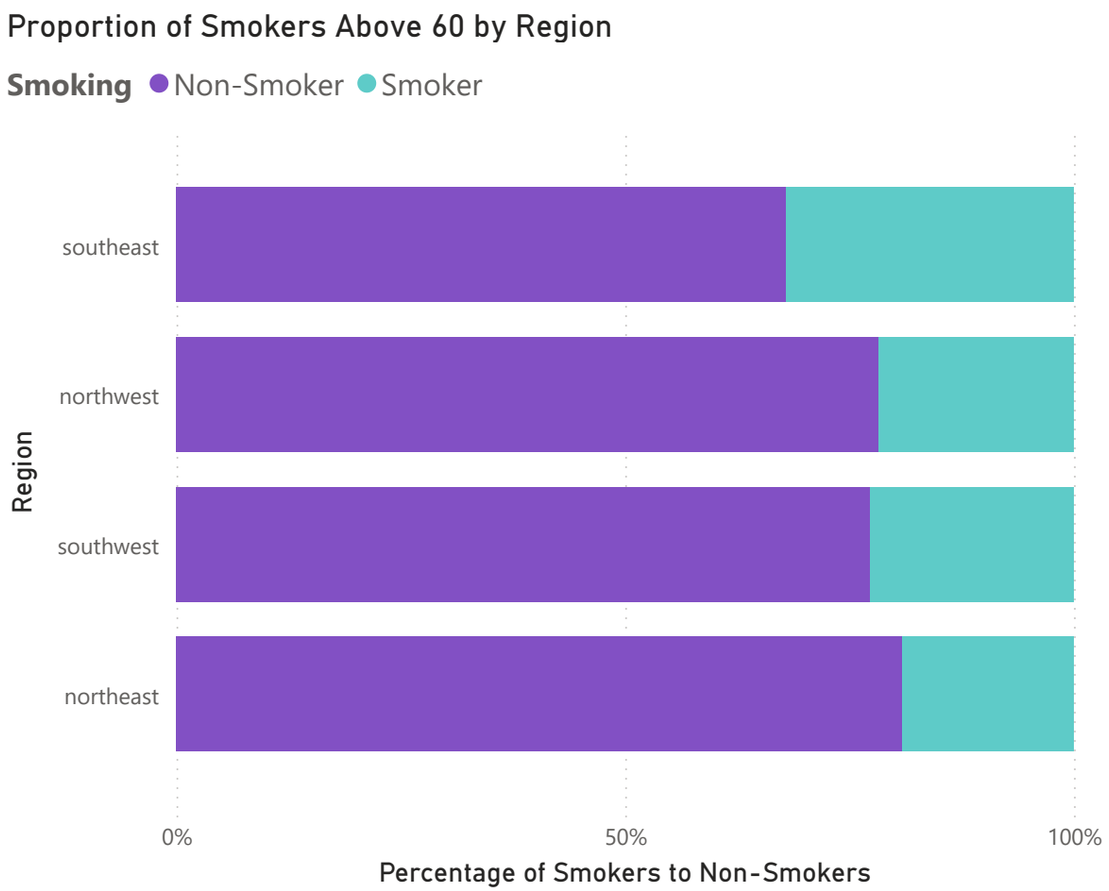


Among smokers, individuals aged 56-60 incur the highest charges, while individuals aged 61-65 incur the highest charges among non-smokers.



Among smokers, average charges are highest in the Southeast (**13.65%** above other regions) and lowest in the Northeast.

Non-smokers face their highest charges in the Northeast and their lowest in the Southwest, with less pronounced regional variation.



The southeast region has the highest proportion of smokers to non-smokers above 60 years old at **32%**, while the northeast is the lowest at 19.05%.

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Conclusion:
Smoking status was the most influential factor associated with high medical insurance charges with a **280%** increase among smokers vs non-smokers, followed by age with **65.83%** increase amongst beneficiaries aged 61-65. The region with the highest proportion of smokers to non-smokers aged above 60 is in the southeast at **32%**.

Recommendations:
Market to young **non-smokers**, aged **18-35** from the **southwest** as they have smallest risk. In addition, organize **preventative health programs** for **smokers** focusing on beneficiaries **above 60** years old in the **southeast** region.